**Project Title:** **Project Design Phase-I - Solution Fit Template** **Team ID:** PNT2022TMIDxxxxxx

Define CS, fit into CC

### 1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

Primary Customers
Healthcare professionals such as doctors and nurses who analyze patient blood pressure data to identify hypertension trends.

Secondary Customers
Medical researchers who need structured datasets for hypertension research and pattern identification.

Tertiary Customers
Healthcare data analysts who work with patient medical records to generate insights for hospitals and research institutions.

### 6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Healthcare professionals and researchers face several constraints that limit their ability to effectively analyze patient data.

Time constraints are one of the major issues. Doctors and healthcare workers often have limited time to manually clean and analyze large datasets.

Technical constraints also exist because many healthcare professionals are **not trained in advanced data analysis tools or programming languages**. They require systems that are simple and easy to use.

### 5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Manual data cleaning using spreadsheets such as Microsoft Excel.

Basic statistical analysis performed manually using spreadsheet formulas.

Using general data analysis tools without healthcare-specific preprocessing.

Manual visualization using separate charting tools.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

### 2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Healthcare professionals need to analyze patient blood pressure records in order to understand hypertension patterns.

They need to identify relationships between symptoms, medication adherence, and blood pressure stages.

They need to detect abnormalities and trends in patient health records quickly.

Researchers need cleaned and structured medical datasets for conducting healthcare studies and research.

Data analysts need to generate visual reports and statistical

### 9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

The root causes of the problems faced by healthcare professionals and researchers when analyzing hypertension datasets are primarily related to **poor data quality and inefficient data processing methods**.

Many patient datasets collected from hospitals and medical records contain **missing values, inconsistent labels, and duplicate entries**. These issues occur because data is often entered manually by different staff members without standardized formats.

Another root cause is the **lack of automated tools for preprocessing healthcare datasets**. Healthcare professionals usually rely on spreadsheets
or manual cleaning methods, which increases the likelihood of human errors.

### 7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Healthcare professionals typically export patient data from hospital systems into spreadsheets.

They manually inspect and clean the data before performing analysis.

Researchers often use statistical tools such as Python or R to perform analysis.

Many healthcare workers rely on charts and visual reports to understand patient health trends.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

### 3. TRIGGERS TR

Healthcare professionals encounter difficulties when analyzing inconsistent or messy patient data.
Researchers often receive datasets that contain missing values, duplicates, or inconsistent labels.
dn cadd

### 4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Before Using the System
Healthcare professionals feel overwhelmed and frustrated when dealing with messy patient datasets.

After Using the System
Healthcare professionals feel confident because the system automatically cleans and organizes patient data.

### 10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

The system performs automated data cleaning to remove duplicates, detect missing values, and standardize categorical labels.

It performs exploratory data analysis to generate statistical summaries and detect patterns in patient blood pressure readings.

The system produces visualizations such as bar charts, histograms, and heatmaps to help healthcare professionals understand hypertension trends.

### 8. CHANNELS of BEHAVIOUR CH

**8.1 ONLINE**
What kind of actions do customers take online? Extract online channels from #7

**8.2 OFFLINE**
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

ONLINE

Medical research platforms and healthcare analytics tools.
Online datasets and research repositories used by researchers.
Software tools such as Python, Jupyter Notebook, and data visualization platforms.
Educational platforms and research communities discussing healthcare data analysis.

OFFLINE
Hospitals and clinics where healthcare professionals analyze patient records.
Medical research institutions conducting healthcare studies.
Academic environments where students and researchers analyze healthcare datasets.

Identify strong TR & EM